

CSR304: ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING – I

B.Tech CSE | AI & Data Engineering Specialisation | Year 2

UNIT I: Introduction to Artificial Intelligence and Machine Learning

Unit I – Topics Covered

This unit covers the foundational landscape of Artificial Intelligence and Machine Learning. Topics include:

- History and evolution of AI and ML
- Distinctions among AI, ML, and Deep Learning
- Applications of AI and ML across domains
- Ethical considerations: bias and fairness
- Key terminologies: algorithms, models, features, labels, parameters, loss functions

1. History and Evolution of AI and Machine Learning

Artificial Intelligence is not a product of the 21st century. Its roots stretch back more than seven decades, shaped by mathematicians, logicians, engineers, and visionaries. Understanding the history helps appreciate why modern AI works the way it does — and what limitations and breakthroughs defined each era.

1.1 The Foundational Era (1940s–1950s)

The intellectual seeds of AI were sown in the 1940s when Alan Turing asked a deceptively simple question in his 1950 paper 'Computing Machinery and Intelligence': Can machines think? He proposed the Turing Test — if a machine's responses are indistinguishable from a human's in a conversation, it can be considered intelligent.

In 1943, Warren McCulloch and Walter Pitts published the first mathematical model of an artificial neuron, demonstrating how nerve cells could theoretically perform logical operations. This laid the groundwork for neural networks. In 1949, Donald Hebb proposed Hebbian Learning — the principle that neurons that fire together, wire together — which remains foundational in modern deep learning.

Example: The Turing Test

Imagine you are chatting online with someone. After a long conversation, you still cannot tell whether you were talking to a human or a computer. According to Turing, that computer has demonstrated intelligence. Modern chatbots like Claude or ChatGPT are the closest realisation of this 70-year-old idea.

1.2 Birth of AI as a Discipline (1956)

The official birth of AI as a scientific discipline is traced to the Dartmouth Conference of 1956, organised by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon. McCarthy coined the term 'Artificial Intelligence' at this conference. The researchers boldly proposed that every aspect of learning and intelligence could be precisely described and simulated by a machine.

This optimism launched the first Golden Age of AI. Early programs like the Logic Theorist (by Allen Newell and Herbert Simon) could prove mathematical theorems. The General Problem Solver attempted to mimic human problem-solving strategies.

1.3 The Perceptron and Early Neural Networks (1958–1969)

Frank Rosenblatt's Perceptron (1958) was the first trainable neural network model. It could classify simple patterns by adjusting weights based on errors. This was revolutionary — a machine learning from examples rather than being explicitly programmed with rules.

However, in 1969, Minsky and Papert published 'Perceptrons', proving that a single-layer perceptron could not solve non-linearly separable problems like XOR. This critique contributed to reduced funding and interest in neural networks, initiating the first AI Winter.

💡 Example: The Perceptron as a Spam Filter

Think of a perceptron as a very early spam filter. You show it thousands of emails labelled 'spam' and 'not spam'. Each time it guesses wrong, it adjusts its internal weights. After enough examples, it learns to detect patterns associated with spam — without you manually coding those rules.

1.4 AI Winters and Revivals (1970s–1990s)

The history of AI is marked by cycles of enthusiasm and disillusionment. An 'AI Winter' refers to a period of drastically reduced funding, public interest, and research output.

First AI Winter (1974–1980)

Early systems failed to scale. The British Lighthill Report (1973) criticised AI for not delivering on its promises. Combinatorial explosion — where the number of possibilities grows exponentially — made it impossible for early computers to handle real-world complexity. Government funding dried up significantly.

Expert Systems Era (1980s)

The 1980s saw a revival through Expert Systems — AI programs that encoded human expertise as IF-THEN rules. Programs like MYCIN (medical diagnosis) and DENDRAL (chemical analysis) demonstrated real utility. Companies invested heavily in rule-based AI.

Second AI Winter (Late 1980s–Early 1990s)

Expert systems were expensive to maintain, brittle, and failed to scale. The AI market collapsed. However, quiet progress continued — particularly in machine learning research.

Connectionist Revival (1986 onwards)

The rediscovery of Backpropagation by Rumelhart, Hinton, and Williams (1986) allowed multi-layer neural networks to be trained effectively. This reignited interest in neural networks and planted the seeds for modern deep learning.

💡 Example: Expert Systems as a Recipe Book

An expert system is like a very detailed recipe book written by a master chef. It has rules like: IF the dish smells burnt THEN lower the heat. Works great for known problems. But if the chef never encountered a problem (like cooking at altitude), the book has no answer. Modern ML doesn't rely on pre-written rules — it discovers them from data.

1.5 The Modern Era: Data-Driven AI (2000s–Present)

The convergence of three key factors in the 2000s–2010s triggered the modern AI revolution:

- **Big Data:** The internet generated unprecedented volumes of labelled data — images, text, transactions.

- Computational Power: GPUs proved highly efficient at the parallel matrix operations central to neural network training.
- Algorithmic Advances: Researchers developed better training techniques, regularisation methods, and architectures.

Key modern milestones:

- 2006: Geoffrey Hinton coined 'Deep Learning' and demonstrated deep belief networks.
- 2009: ImageNet dataset released — 14 million labelled images for computer vision benchmarking.
- 2012: AlexNet wins the ImageNet competition, showcasing deep CNNs with dramatically lower error rates.
- 2016: AlphaGo defeats world Go champion Lee Sedol through reinforcement learning.
- 2017: The Transformer architecture ('Attention is All You Need') revolutionises NLP.
- 2022–2025: Large Language Models (GPT-4, Claude, Gemini) demonstrate remarkable general-purpose capabilities.

AI Timeline at a Glance

1943 → McCulloch-Pitts neuron model | 1950 → Turing Test
 1956 → Dartmouth Conference; AI coined | 1958 → Rosenblatt's Perceptron
 1974–1980 → First AI Winter | 1980s → Expert Systems boom
 1986 → Backpropagation rediscovered | 1990s → SVM, statistical ML
 2006 → Deep Learning (Hinton) | 2012 → AlexNet; deep CNN revolution
 2016 → AlphaGo | 2017 → Transformer | 2022+ → Generative AI, LLMs

2. Distinctions Among AI, ML, and Deep Learning

One of the most commonly confused aspects in the field is the relationship between Artificial Intelligence, Machine Learning, and Deep Learning. These three terms are related but represent distinct concepts arranged in a nested hierarchy.

2.1 Artificial Intelligence (AI)

Artificial Intelligence is the broadest concept. It refers to the simulation of human-like intelligence by machines — the ability to perceive the environment, reason, learn, and take purposeful action. AI can be rule-based (expert systems) or learning-based (machine learning).

Definition: Artificial Intelligence

AI is the science and engineering of creating intelligent machines — systems that perceive, reason, learn, and act to achieve goals.

AI can be classified by capability:

- **Narrow AI:** Narrow AI (Weak AI): Designed for a specific task. All current AI systems are narrow — chess engines, facial recognition, voice assistants.
- **General AI:** General AI (AGI): A hypothetical AI with human-level intelligence across all domains. Does not yet exist.
- **Superintelligent AI:** Superintelligent AI: A hypothetical AI that surpasses human intelligence. Subject of significant AI safety research.

2.2 Machine Learning (ML)

Machine Learning is a subset of AI. Rather than being explicitly programmed with rules, an ML system learns patterns from data. The programmer does not tell the system what to look for — the system discovers patterns through exposure to examples.

Definition: Machine Learning

Machine Learning is a subset of AI that enables systems to learn and improve from experience (data) without being explicitly programmed for every task.

The key distinction between traditional programming and machine learning:

Traditional Programming	Machine Learning
Input: Data + Hand-coded Rules	Input: Data + Correct Answers (Labels)
Output: Answers/Decisions	Output: A Learned Model (Rules)
Rules written by humans	Rules discovered from data automatically

Fails when rules are complex or change

Adapts by retraining on new data

💡 Example: Spam Detection: Traditional vs ML

Traditional: A programmer writes rules — 'If email contains FREE in caps, mark as spam.' This breaks when spammers change tactics.

ML: Show the system thousands of pre-labelled emails. It learns complex patterns automatically — even recognising new spam tactics that resemble old ones.

2.3 Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with many layers. Each layer automatically learns increasingly abstract representations of the input data. While traditional ML requires feature engineering (humans manually selecting meaningful features), deep learning can learn features directly from raw data given enough data and compute.

Definition: Deep Learning

Deep Learning is a subset of ML using multi-layered artificial neural networks that automatically learn hierarchical feature representations from raw data.

Why 'deep'? In image recognition, for example:

- Layer 1 might detect edges and corners from raw pixels.
- Layer 2 combines edges to detect shapes like circles and rectangles.
- Layer 3 combines shapes to detect parts like eyes, noses, and ears.
- Final layers combine everything to recognise a face.

💡 Example: Learning to Recognise a Mango

Traditional ML: You manually define features — colour = yellow/green, shape = oval, texture = smooth. The model classifies based on your chosen features.

Deep Learning: You just show the network thousands of mango images. The first layers detect colour gradients and curves. Deeper layers combine these into the characteristic mango shape — all learned automatically, no manual features needed.

2.4 The Nested Hierarchy

- All Deep Learning is Machine Learning, but not all Machine Learning uses deep neural networks.
- All Machine Learning is AI, but not all AI uses ML — rule-based expert systems are AI but not ML.
- Deep Learning is the most constrained subset, requiring large datasets and significant compute power.

The Nested Structure

[Artificial Intelligence – the broadest concept]

[Machine Learning – learning from data]

[Deep Learning – multi-layer neural nets]

3. Applications of AI and ML Across Domains

AI and ML have permeated nearly every major industry. Understanding these applications contextualises the algorithms you will learn throughout this course and demonstrates the real-world impact of the techniques covered.

3.1 Healthcare and Medical Sciences

AI is transforming medicine from diagnosis to drug discovery. ML models analyse medical images (X-rays, MRIs, CT scans) and detect conditions like tumours, fractures, and diabetic retinopathy with accuracy matching specialist doctors.

- **Medical Imaging:** Deep CNNs like DenseNet detect pneumonia in chest X-rays with high accuracy. FDA has approved over 500 AI-based medical imaging tools.
- **Drug Discovery:** ML predicts how molecules interact with biological targets, dramatically accelerating new drug identification.
- **Personalised Medicine:** Algorithms analyse patient genomic data to recommend treatments tailored to individual genetic profiles.
- **Clinical Decision Support:** NLP systems extract insights from unstructured clinical notes to assist doctors.

Example: AlphaFold – Greatest AI Breakthrough

Protein folding — predicting 3D protein structure from amino acid sequences — was a 50-year unsolved biological challenge. DeepMind's AlphaFold (2020) solved it with remarkable accuracy, potentially revolutionising drug design and disease research.

3.2 Finance and Banking

- **Fraud Detection:** ML models analyse thousands of transaction features in real-time to flag suspicious activity. Credit card companies process millions of transactions daily.
- **Credit Scoring:** ML assesses loan applicants using diverse features beyond traditional credit history.
- **Algorithmic Trading:** Quantitative hedge funds use ML to identify price patterns and execute trades in milliseconds.
- **Risk Management:** Models forecast market volatility and assess portfolio risk under various scenarios.

3.3 Transportation and Autonomous Vehicles

- **Self-Driving Cars:** Tesla, Waymo, and others deploy deep learning models processing camera, LiDAR, and radar data to navigate roads.
- **Traffic Optimisation:** Google Maps uses ML to predict traffic conditions and suggest optimal routes in real time.
- **Predictive Maintenance:** Airlines use ML to predict component failures before they occur, preventing costly delays.

3.4 Natural Language Processing (NLP)

- Machine Translation: Google Translate uses transformer models to translate between 100+ languages.
- Search Engines: Google's BERT model understands semantic intent behind queries rather than just matching keywords.
- Voice Assistants: Alexa, Siri, and Google Assistant use speech-to-text, NLP, and text-to-speech pipelines.
- Text Generation: LLMs like Claude and ChatGPT summarise documents, write code, answer questions, and more.

3.5 Recommendation Systems

Recommendation systems are arguably the most commercially impactful ML application, driving enormous revenue.

- Amazon: Estimated 35% of revenue attributed to its recommendation engine.
- Netflix: The recommendation algorithm saves an estimated \$1 billion per year in customer retention.
- YouTube: Recommendations drive approximately 70% of total watch time.
- Spotify: Curates personalised playlists (Discover Weekly) using collaborative filtering and audio analysis.

Example: How Netflix Recommendations Work

Netflix uses collaborative filtering: 'Users who watched what you watched also liked these.' Combined with your own watching patterns (when you pause, rewind, abandon), it builds a personalised taste profile. The same movie is even shown with different thumbnail images to different users based on predicted preferences!

3.6 Agriculture and Environment

- Precision Agriculture: ML analyses satellite imagery and sensor data to optimise irrigation, detect crop diseases, and predict yields.
- Climate Modelling: Deep learning accelerates weather forecasting significantly beyond traditional physics-based models.
- Wildlife Conservation: Computer vision identifies endangered species in camera trap images automatically.

3.7 Education

- Adaptive Learning: Platforms like Khan Academy adapt content difficulty based on individual student performance.
- Automated Essay Grading: NLP models provide consistent, scalable feedback on student writing.
- Student Performance Prediction: Models identify at-risk students early so institutions can intervene.

4. Ethical Considerations in AI: Bias and Fairness

As AI systems increasingly make or assist in consequential decisions — who gets a loan, who gets hired, who gets bail, which patients receive treatment — the ethical dimensions become critically important. This is not merely a technical concern but a societal and human rights issue.

4.1 What is Algorithmic Bias?

Algorithmic bias refers to systematic and unfair discrimination by an AI system against particular groups of people, based on characteristics like race, gender, age, religion, nationality, or socioeconomic status. Crucially, bias in AI often arises not from malicious intent but from flawed data, flawed problem framing, or flawed evaluation criteria.

Key Insight

AI bias is fundamentally a social problem that manifests technically. Even mathematically 'correct' models can perpetuate historical injustice if trained on historically biased data.

4.2 Sources of Bias

4.2.1 Historical Bias

When training data reflects historical inequalities, the model learns and perpetuates those inequalities. If historical hiring data shows only men were hired for engineering roles (due to past discrimination), a model trained on this data will prefer male candidates.

💡 Example: Amazon's Hiring Algorithm (2018)

Amazon built an AI recruiting tool trained on 10 years of successful hires — predominantly male. The model learned to penalise resumes containing words like 'women's' (e.g., 'Women's Chess Club'). Amazon scrapped the tool when this bias was discovered. Textbook historical bias.

4.2.2 Representation Bias (Sampling Bias)

When the training dataset does not represent the population the model will be deployed on. If a facial recognition system is trained predominantly on lighter-skinned faces, it performs poorly on darker-skinned individuals — with severe potential consequences in law enforcement.

💡 Example: Facial Recognition Accuracy Disparities

MIT's Joy Buolamwini (2018 Gender Shades study) found commercial facial recognition systems had error rates below 1% for lighter-skinned men, but up to 34% for darker-skinned women. Cause: training datasets were predominantly composed of lighter-skinned male faces.

4.2.3 Measurement Bias

When features or labels are measured differently for different groups. Using arrest rates as a proxy for crime rates introduces bias because policing is not uniform — over-policed communities appear to have higher crime in data even if actual crime rates are similar.

4.2.4 Aggregation Bias

When a model is trained on combined data from multiple groups but the relationships differ across groups. A diabetes management model trained on a combined population may perform poorly for specific ethnic groups with different physiological relationships.

4.2.5 Deployment Bias

When a model is used in a context different from where it was developed. A model developed on urban hospital data may perform poorly in rural hospital settings.

4.3 Fairness Definitions in Machine Learning

Defining 'fairness' is surprisingly complex. Multiple mathematical definitions exist, and they can be mutually incompatible — satisfying one may violate another.

Fairness Concept	Meaning
Demographic Parity	Equal positive prediction rates across groups. E.g., loan approvals should be equally common across racial groups.
Equal Opportunity	Equal true positive rates. Among qualified applicants, the model correctly identifies them at equal rates regardless of group.
Individual Fairness	Similar individuals should be treated similarly. Focuses on individual cases rather than group membership.
Calibration	If a model assigns 70% risk score, approximately 70% of those cases should actually be high risk, across all groups.

4.4 Transparency, Accountability, and Explainability

- **Transparency:** Transparency: Users and affected parties should know when and how an AI system is making decisions about them. Black-box models are problematic in high-stakes domains like criminal justice or healthcare.
- **Accountability:** Accountability: Someone must be responsible when an AI system causes harm — developers, deploying organisations, or regulators.
- **Explainability:** Explainability (XAI): AI systems should provide meaningful explanations for decisions. Techniques like LIME, SHAP, and attention visualisation help make model decisions interpretable.
- **Privacy:** Privacy: AI systems require vast personal data. Techniques like differential privacy and federated learning protect individual privacy.

Example: COMPAS Recidivism Algorithm

COMPAS is used in US courts to predict whether a criminal will reoffend, influencing sentencing. A 2016 ProPublica investigation found it was twice as likely to falsely flag Black defendants as high risk compared to white defendants. The inputs included neighbourhood and social factors correlated with race — a direct example of how AI bias translates into discriminatory outcomes.

4.5 Mitigating Bias: Technical Approaches

- Pre-processing: Resampling, reweighting, or transforming training data to reduce disparities before training.
- In-processing: Modifying the learning algorithm to incorporate fairness constraints during training.
- Post-processing: Adjusting model outputs or decision thresholds differently for different groups after training.
- Diverse Teams: Diversity among AI developers helps identify blind spots that homogeneous teams miss.
- Regular Auditing: Continuously monitoring deployed models for disparate impact and performance drift.

5. Key Terminologies in AI and Machine Learning

A solid understanding of core ML terminology is essential for reading research papers, understanding documentation, and communicating with other practitioners. This section covers the most fundamental concepts you will encounter throughout this course.

5.1 Data

Data is the raw material of machine learning. Without data, ML models cannot be trained. Data can take many forms:

- **Structured Data:** Organised in tables with defined columns (CSV files, database tables with numerical/categorical values).
- **Unstructured Data:** No predefined format — text documents, images, audio files, videos.
- **Semi-Structured Data:** Some organisation but not rigid schema — HTML, XML, JSON.
- **Time-Series Data:** Sequential data indexed by time — stock prices, sensor readings, weather data.

Dataset	A collection of data examples used to train, validate, or test a machine learning model.
Training Set	The portion of data used to train (fit) the model. The model sees these examples and adjusts its parameters.
Validation Set	Data used to tune model hyperparameters and make design decisions. The model does not train on this.
Test Set	Data held back until final evaluation. Measures real-world generalisation. Never used during training or tuning.

5.2 Features and Labels

These are the two most fundamental concepts in supervised learning.

Features (X)	The input variables — measurable properties or attributes of the data used to make predictions. Also called predictors, attributes, or independent variables.
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Problem Domain	Features (X)	Label (Y)
House Price Prediction	Area, bedrooms, location, age	Price (in lakhs)
Email Spam Detection	Word frequencies, sender, links count	Spam / Not Spam

Disease Diagnosis	Age, blood pressure, glucose, BMI	Diabetic / Not Diabetic
Image Classification	Pixel values (R, G, B channels)	Cat / Dog / Car / etc.

Labels (Y)	The output variable — the 'answer' the model is trying to predict. Also called target variable, response variable, or dependent variable.
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Feature Engineering	Using domain knowledge to create, transform, or select features that improve model performance. Often considered more art than science.
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5.3 Algorithms and Models

Algorithm	A set of mathematical procedures the computer follows to learn patterns from data. Examples: Linear Regression, Decision Tree, k-Means, Neural Network.
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Model	The output of running a learning algorithm on training data. Captures learned patterns and makes predictions on new data.
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Example: Algorithm vs Model Analogy

The Linear Regression algorithm is like the process of 'draw the best fitting line through these points.' The resulting model is the actual line — a specific equation with specific slope and intercept values — learned from your specific data. The same algorithm applied to a different dataset produces a different model.

5.4 Training and Generalisation

Training	The process of adjusting model parameters so the model learns patterns from training data. Also called 'fitting' the model.
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Prediction (Inference)	Using a trained model to generate outputs for new, unseen inputs.
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Generalisation	A model's ability to perform well on new, unseen data — not just the data it was trained on. The ultimate goal of ML.
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Overfitting

When a model learns training data too well — including noise and irrelevant patterns — resulting in poor performance on new data. Like memorising answers without understanding concepts.

Underfitting

When a model is too simple to capture underlying patterns, performing poorly on both training and test data. Like studying too little.

💡 Example: Overfitting vs Underfitting in Exam Prep

Underfitting: You studied a few concepts superficially. You perform poorly even on familiar question types. The model is too simple.

Overfitting: You memorised textbook questions word-for-word. You score perfectly on practice tests but fail the actual exam with slightly different questions. The model memorised the training set.

Good Fit: You understood the concepts. You can answer both familiar and novel questions well. This is generalisation.

5.5 Parameters and Hyperparameters

Parameters

Internal values learned by a model during training. For example: weights and biases in a neural network, or slope and intercept in linear regression.

Hyperparameters

Configuration settings chosen before training begins. Examples: learning rate, number of trees in a random forest, depth of a decision tree.

💡 Example: Parameters vs Hyperparameters in Baking

Hyperparameters are like oven settings you choose before baking — temperature (200°C), time (30 mins), pan size. Parameters are like the exact texture and colour your cake develops during baking — determined by the process. You set the oven; the baking process determines the cake's specific properties.

5.6 Loss Function and Optimisation

Loss Function

Measures how far off model predictions are from the true labels. Training minimises this function. Also called cost function or objective function.

Gradient Descent	An optimisation algorithm that iteratively adjusts model parameters in the direction that reduces the loss. Like rolling a ball downhill to find the valley floor.
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Learning Rate	A hyperparameter controlling the step size in gradient descent. Too large: overshoots the minimum. Too small: training is very slow.
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5.7 Evaluation Metrics

Accuracy	Proportion of correct predictions. Formula: $\text{Correct Predictions} / \text{Total Predictions}$. Misleading for imbalanced datasets.
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Precision	Of all positive predictions made, what fraction was actually positive? High precision = few false alarms. Critical when false positives are costly.
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Recall (Sensitivity)	Of all actual positives, what fraction did the model correctly identify? Critical when false negatives are costly (e.g., disease detection).
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F1 Score	Harmonic mean of Precision and Recall. $F1 = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$. Useful for imbalanced classes.
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Mean Squared Error (MSE)	Average squared difference between predictions and actual values. Used for regression. Penalises large errors heavily.
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5.8 Other Important Terms

Pipeline	A sequence of data processing and model training steps chained together: cleaning → scaling → training → evaluation.
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Curse of Dimensionality	As the number of features grows, data becomes increasingly sparse, making it harder to find meaningful patterns and requiring exponentially more data.
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Batch vs Online Learning	Batch learning trains on the full dataset at once. Online learning updates the model as new data arrives, one sample or small batch at a time.
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Inference

Using a trained model in production to make predictions. The deployment phase of an ML system.

Unit I: Sample Questions (20 Questions)

Exam Preparation Note

These 20 questions cover all Unit I topics — from 2-mark definitions to 10-mark analytical questions. Practice writing structured answers with clear definitions, theory, and relatable examples.

Part A: Short Answer Questions (2–3 marks)

- Q1.** Define Artificial Intelligence. How is it different from traditional programming? Give one example.
- Q2.** What is the Turing Test? Who proposed it and in which year? Is it still relevant today?
- Q3.** Distinguish between features and labels with an example from a medical diagnosis scenario.
- Q4.** What is the difference between a training set and a test set? Why must they be kept completely separate?
- Q5.** Define overfitting and underfitting. Which is more dangerous in a medical diagnosis AI system? Justify.
- Q6.** What are hyperparameters? Give two examples and explain how they differ from model parameters.
- Q7.** Define algorithmic bias. Name two sources of bias that can affect an AI system.
- Q8.** Explain the concept of a loss function and why minimising it is the core objective of model training.

Part B: Medium Answer Questions (5 marks)

- Q9.** Explain the relationship between Artificial Intelligence, Machine Learning, and Deep Learning using the nested hierarchy concept. Give an example of when you would use each approach.
- Q10.** Describe the two AI Winters in history. What caused each, and what key lessons did the AI research community learn?
- Q11.** Explain representation bias in AI. Describe the Gender Shades study and what it revealed. What steps can developers take to mitigate representation bias?
- Q12.** What is the difference between an algorithm and a model? Using house price prediction as an example, trace how an algorithm is applied to data to produce a model.
- Q13.** Describe any four real-world application domains of Machine Learning. For each, mention one specific application and what type of ML it uses.
- Q14.** Compare and contrast batch learning and online learning. Provide a specific use case where each approach is more suitable.

Part C: Long Answer / Analytical Questions (10 marks)

Q15. Write a detailed note on the history and evolution of Artificial Intelligence from the 1940s to the present. Cover: foundational contributions (McCulloch-Pitts, Turing, Rosenblatt), expert systems, the two AI winters, the deep learning revolution, and factors driving modern AI.

Q16. Explain the three mathematical definitions of ML fairness: Demographic Parity, Equal Opportunity, and Calibration. Why can these definitions conflict with each other? Illustrate with the COMPAS criminal justice system. What does this reveal about the social nature of 'fairness' in AI?

Q17. Describe the role of Feature Engineering in Machine Learning. What is the Curse of Dimensionality? How does Deep Learning reduce the need for manual feature engineering, and what trade-offs does this create compared to classical ML?

Q18. Compare rule-based AI (expert systems) with modern machine learning across: development approach, scalability, maintenance requirements, failure modes, and appropriate use cases. When would you still prefer a rule-based system over an ML model?

Q19. Define and explain the following terms with practical examples from relevant domains: (a) Generalisation and Overfitting, (b) Loss Function and Gradient Descent, (c) Precision and Recall, (d) Feature Engineering, (e) Dimensionality and the Curse of Dimensionality.

Q20. You are tasked with building an AI system to screen resumes for a technology company. Identify three types of bias that could affect this system. For each: define the bias type, explain how it manifests in this specific scenario, and suggest a technical and/or procedural mitigation strategy. Conclude with a reflection on whether completely bias-free AI is achievable.